# NegativeHours: Milestone 2 summary

Corsin Hidber · HSLU MLOps HS26 · Repo: `CorsinAI/MLOPS-Project-HSLU-HS-26`

## What was achieved

The feature pipeline runs end to end. GitHub Actions fetches SMARD data and Open-Meteo forecasts, builds each row as known at 10:00 Berlin on D-1, validates it and upserts it to Hopsworks. Live runs and backfills share the same code. Training and inference remain out of scope.

| Proof | Result | GitHub Actions run |
|---|---|---|
| Hopsworks round trip | Login, insert, upsert in place, read and delete all passed | 41913408675 |
| Daily pipeline, manual start | Success: 216 feature and 216 label rows, no failed source | 41914046157 |
| Scheduled daily run | Cron is live on `main`; first slots are 01:17 and 05:17 UTC on 05.11. Not yet proven | Pending |
| Full backfill and stored-data report | Success in 243 s; all checks passed | 41915601205 |
| CI on `main` | Lock check, ruff, 128 tests and Docker build passed | 41915591120 |

**Stored data, read back from Hopsworks after the backfill:**

- **Rows:** 24,960 in `neghour_features` and 24,960 in `neghour_labels`, covering 2024-01-01 to 2026-11-05, with 0 duplicate UTC hours.
- **Clock changes:** spring days 2024-03-31, 2025-03-30 and 2026-03-29 have 23 rows. Autumn days 2024-10-27, 2025-10-26 and 2026-10-25 have 25.
- **Leakage:** 0 rows have any `available_at_*` after the cut-off, which the report recomputes from `delivery_day`. The closest value is 3 hours before it.
- **Gaps:** `wind_100m` and the proxy are empty for 1,135 hours, to 2024-02-17 05:00 UTC. Irradiance, cloud cover and temperature are empty for 469 hours, to 2024-01-20 11:00 UTC. Both match the start of archived forecasts. Price and actuals have no gaps. `n_neg_quarters` starts at the 15-minute switch, as designed.
- **Negative hours:** 457 of 8,784 in 2024 (5.20 %), 573 of 8,760 in 2025 (6.54 %), 462 of 7,416 in 2026 so far (6.23 %). The total is 1,492 (5.98 %), identical to the dry run.
- **Hopsworks jobs:** 0 executions. Two `compute_stats` job definitions exist but have never run, because statistics are off and inserts write Delta tables directly.

## Key decisions

- **Label:** `y_neg` = hourly mean price < 0 from SMARD filter 4169. It is the one definition valid across the whole dataset. `n_neg_quarters` is stored so a quarter-hour label can be tested later.
- **Vintage:** `_previous_day2` forecasts in both backfill and live, so training and serving see forecasts of the same age.
- **Time:** storage in UTC, keyed by UTC hour; delivery days and the cut-off in Europe/Berlin. Rows are selected by timestamp, so a late cron run writes the same features.
- **Availability:** rule-based, because SMARD publishes no per-row timestamp.
- **Late data:** NaN plus a `missing_*` flag, never carried forward.

**Known limits:** SMARD revisions and the stability of `_previous_day2` values between runs are not measured. The 6-hour publication margins are assumptions. All decisions are in `docs/ms2_decisions.md`.

## What to look at first

1. `tests/test_leakage.py`: changes or removes data published after the cut-off and requires identical rows.
2. `src/neghour/availability.py`: all publication rules in one place.
3. `scripts/check_store_quality.py`, and its output in backfill run 41915601205.
4. `docs/proposal_addendum.md`: every change from the proposal, with its review point.

## Response to the review points

The editorial uses E2 twice, so its two uses appear here as E2a and E2b.

| ID | Point | Status | Response | Commit or file |
|---|---|---|---|---|
| A1 | Same weather vintage in backfill and live | Changed | `_previous_day2` in both. Valid-time availability is T − 48 h + 6 h. | 304cb25, `sources/open_meteo.py` |
| A2 | Europe/Berlin everywhere; CET/CEST in UTC cron | Changed | UTC storage, Berlin days. 23 and 25 hour days tested and confirmed in storage. | 49ebba4, b9f0ddc, `timeutils.py` |
| A3 | Test that fails on data after the cut-off | Changed | Checked before each write, by tests, and on the full stored table: 0 late rows. | 245c48b, aa75bc7, 689eb2f |
| B1 | Start before 10:00; define late and missing data | Changed | Cron at 01:17 and 05:17 UTC, since runs started 4–9 h late. Policy in the README. | c91f563, dc8bdbf |
| B2 | Weighted holiday share | Changed | `holiday_share` by state population, next to `is_holiday`. | c0d414d, `features/calendar.py` |
| E1 | Hourly label after the 15-minute switch | Changed | Hourly mean, justified by one label across all splits; quarter count stored. | 6603e83, 6f812e5, 85bbfe8 |
| E2a | Resolve the confusing proposal points | Changed | Addendum lists each change; the submitted proposal is untouched. | 5d9084c, `proposal_addendum.md` |
| E2a-i | Classification or price prediction | Changed | Classification only. | 6f812e5, `proposal_addendum.md` |
| E2a-ii | Region covered by each input | Changed | Four points, averaged per variable; reasoning in the README. | 304cb25, 72e3906 |
| E2a-iii | Seasonality within year, day and week | Partly | Features built and argued. Modelling follows in MS3. | c0d414d, `README.md` |
| E2a-iv | Upsampling causes training-serving skew | Partly | Upsampling rejected in writing. `scale_pos_weight` follows in MS3. | 021b736, `README.md` |
| E2b | Originality against KTH SE3 regressors | Changed | README line added. | 021b736, `README.md` |
| E3 | Rolling PR-AUC/F1 and drift check | Not yet | Needs a model, so it is planned after MS3 training. | None yet |
| E4 | Commit in small steps | Changed | Over 30 focused commits, one pull request per phase. | PRs 1, 2, 4–9 |